

An Exploration of Softmax Alternatives Belonging to the Spherical Loss Family

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Motivation

- ▶ Softmax is standard for multi-class classification, but this choice is somewhat arbitrary as its superiority is not theoretically justified.
- ▶ The spherical loss family allows efficient gradient updates in $O(d^2)$ instead of $O(d \times D)$.
- ▶ In their paper, Alexandre de Brebisson and Pascal Vincent explore spherical losses as alternatives to log-softmax for classification tasks.

Baseline: Log-Softmax

$$L(\mathbf{o}, c) = -\log \frac{e^{o_c}}{\sum_{k=1}^D e^{o_k}} = -o_c + \log \sum_{k=1}^D e^{o_k}$$

- Standard in classification, but computationally expensive for large D : gradient updates require $O(D \times d)$.

Spherical Loss Family: Definition

A loss belongs to the spherical family if it depends only on:

$$s = \sum_i o_i, \quad q = \|\mathbf{o}\|^2, \quad o_c \text{ and } y_c \text{ for target class } c :$$

$$\mathcal{L} = \mathcal{L}(s, q, o_c, y_c)$$

- ▶ Enables efficient gradient computation¹ in $O(d^2)$.
- ▶ A classic example is MSE.

¹*Vincent, Pascal, de Brebisson, Alexandre, and Bouthillier, Xavier.* Efficient exact gradient update for training deep networks with very large sparse targets. NIPS, 2015.

Spherical Upper Bounds of Log-Softmax

- ▶ Derived from Bouchard (2007)² upper bound of log-sum-exp.
- ▶ Form:

$$L \leq \left(-\frac{(D-2)^2}{16D} \frac{1}{\lambda(\xi)} - \frac{D}{2} \xi - D\lambda(\xi)\xi^2 + \right. \\ \left. + D \log(1 + e^\xi) + \frac{1}{D} s + \left(q - \frac{s^2}{D} \right) \lambda(\xi) - o_c \right), \\ \lambda(\xi) = \frac{1}{2\xi} \left(\frac{1}{1 + e^{-\xi}} - \frac{1}{2} \right)$$

- ▶ Disadvantages:
 - ▶ Bound minimization did not improve performance.
 - ▶ Hyperparameter ξ sensitive and hard to tune.
 - ▶ Failed to outperform even initial model in language tasks.

²Bouchard, Guillaume. Efficient bounds for the softmax function and applications to approximate inference in hybrid models (2007).

Spherical Softmax Loss

$$f_{\text{sph_soft}}(\mathbf{o})_k = \frac{o_k^2 + \varepsilon}{\sum_i (o_i^2 + \varepsilon)}$$

$$L_{\text{log_sph_soft}} = -\log f_{\text{sph_soft}}(\mathbf{o})_c$$

- ▶ Invariant to scaling of $\mathbf{o}$.
- ▶ Requires careful tuning of ε for numerical stability.
- ▶ Even function (ignores sign of o_k).

Taylor Softmax Loss

Based on Taylor expansion of $\exp(x) \approx 1 + x + \frac{1}{2}x^2$:

$$f_{\text{tay_soft}}(\mathbf{o})_k = \frac{1 + o_k + \frac{1}{2}o_k^2}{\sum_i (1 + o_i + \frac{1}{2}o_i^2)}$$

$$L_{\text{log_tay_soft}} = -\log f_{\text{tay_soft}}(\mathbf{o})_c$$

- ▶ No extra hyperparameter needed.
- ▶ Slightly asymmetric around zero.
- ▶ Numerically stable.

Results: MNIST and CIFAR-10

Loss	MNIST Error	CIFAR-10 Error
Log-Softmax	0.812%	8.52%
Log-Taylor Softmax	0.785%	8.07%
Log-Spherical Softmax	0.828%	8.37%

- ▶ Taylor softmax outperforms log-softmax on low-dimensional outputs.
- ▶ Spherical losses train faster and achieve lower error.

Results: CIFAR-100 and Language Modeling

Loss	CIFAR-100 Error	PennTree Perplexity
Log-Softmax	32.4%	126.7
Log-Taylor Softmax	33.1%	147.2
Log-Spherical Softmax	33.1%	149.2

- ▶ Log-softmax performs better as output dimension increases.
- ▶ Exponential in softmax may better handle high-dimensional competition.
- ▶ Spherical losses struggle in language tasks with large vocabularies.

Key Takeaways

- ▶ Spherical losses (especially Taylor softmax) can outperform log-softmax on small-output tasks (e.g., MNIST, CIFAR-10).
- ▶ For high-dimensional outputs (CIFAR-100, language modeling), log-softmax remains superior.
- ▶ Spherical losses enable efficient training but may lack the discriminative power of softmax in complex scenarios.
- ▶ Spherical alternatives are worth considering for specific applications with low output dimension or where efficiency is key.